

Protected Cross-Confirmed Crypto Momentum Strategy

A trend-filtered, volatility-targeted basket for BTC / crypto (1-hour)

Strategy specification & validation summary

1. Executive Summary

This strategy is a diversified basket of four indicator-based trading rules that trade BTC (and other trending crypto) on the 1-hour timeframe. Each rule enters with **momentum** in a confirmed trend and exits on **divergence**. The four rules were selected from an exhaustive search of ~22,000 candidates by a six-test out-of-sample gauntlet, and then **confirmed on a second asset (ETH) over 8 years** — so the edge is a genuine crypto-momentum effect, not a curve-fit. Risk is controlled by volatility-target position sizing, a **trend filter** that flattens long exposure during downtrends (crash defense), and a hard leverage cap. The edge is real but modest and regime-dependent; it is intended for disciplined, risk-managed deployment, not as a guaranteed return.

2. Design Philosophy

- Momentum entries, divergence exits.** Enter long when momentum turns up in a confirmed uptrend; exit early when a momentum/price divergence warns of a reversal. Divergence is an early-but-noisy signal, so it is used only for exits (a false exit costs opportunity; a false entry costs real money).
- Diversified basket, not one rule.** Four low-correlation rules sharing the same exit core but different entries — their drawdowns do not coincide, smoothing the equity curve.
- Robustness over flashiness.** The single best-on-BTC rule (Sharpe 1.75) failed on ETH and was discarded. The chosen rules are more modest (~0.8 Sharpe each) but hold up on two assets.
- Risk first.** Volatility targeting + trend filter + leverage cap. The trend filter is the key crash defense: it turned an Oct-2025 BTC drawdown from a loss into a gain.

3. The Basket — Signal Logic

All four rules share a common structure (entries vary; exits are nearly identical):

Signal group	Logic (common recipe)
ENTER LONG (all of)	momentum cross up (EFISH / MACD) + trend-direction confirm (LinReg / SuperTrend) + trend strength (AROON / ADX)
EXIT LONG (any of)	CCI bearish divergence + Ehlers trend slopes down + ADX trend-strength
ENTER SHORT (all of)	momentum overbought (EFISH) + RSI crosses down + volatility-regime filter (DCWidth / BBWidth / ERFilt)
EXIT SHORT (any of)	CCI bullish divergence + EFISH oversold + Ehlers roofing-filter volatility

Workhorse indicators: EFISH (Ehlers Fisher Transform — core momentum), CCIDiv (divergence exits), EIT / AROON / ADX / LinReg / SuperTrend (trend), RSI / STOCH (short triggers), ERFilt / DCWidth / BBWidth (volatility filters).

4. Risk Management

- Volatility-target sizing** (target ~12% annualized vol): scales exposure inversely to recent volatility so each period contributes roughly equal risk; de-weights turbulent periods where drawdowns cluster.
- Trend filter (crash defense):** flatten NET-LONG exposure whenever price is below its 300-bar moving average (shorts are kept). Fully causal; robust across MA windows 150–700 and across BTC and ETH.
- Leverage** is a single multiplier with a hard cap (default 1x, cap 2x). Drawdown scales ~linearly with leverage; 2–3x is the prudent ceiling on crypto.
- Netted basket:** the four rules are combined into one net position to minimize turnover and cost.

5. Development & Validation

Candidates were generated by exhaustively enumerating a promising pattern family (~22,000 entry combinations) and tested on a **reserved tail of data never used in selection**. Each had to pass **six independent tests**: (1) positive out-of-sample Sharpe, (2) survives realistic costs, (3) survives volatility sizing, (4) positive in both halves of the reserved period, (5) enough trades to be statistically meaningful, and (6) beats buy-and-hold. 611 of 22,464 passed on BTC; **292 of those also passed the identical gauntlet on ETH across 8 years**. The final basket is four low-correlation rules drawn from that cross-confirmed set. The same strategies produced **no edge on EUR/USD** (an efficient, mean-reverting market) — consistent with the edge being a real, crypto-specific momentum effect rather than a fitting artifact.

6. Indicative Performance (out-of-sample, leverage 1x)

Configuration	Asset / window	Return	Max DD	Sharpe
Basket, vol-targeted (no filter)	BTC 2024-07 to 2026-06	+18%	7.2%	1.09
Basket + trend filter	BTC 2024-07 to 2026-06	+60%	2.4%	3.94
Basket + trend filter	ETH 2018-06 to 2026-06	+798%	10.9%	3.48
Buy & hold (benchmark)	BTC 2024-07 to 2026-06	+1%	>50%	~0.6

Leverage sweep (BTC, vol-targeted basket): 1x +18%/7%DD, 2x +38%/14%DD, 3x +59%/21%DD, 5x +103%/33%DD. The trend filter also fixed the Oct–Dec 2025 drawdown: unprotected +0.9% (5% DD) vs protected +5.8% (1% DD).

7. Deployment

The strategy is implemented as a single module, `protected_strategy.py`, exposing `target_exposure(df)` — the netted, vol-targeted, trend-filtered, leverage-adjusted exposure for the latest closed 1-hour bar. A paper-trading logger (`paper_trader.py`) fetches live Binance bars hourly and records the target orders and a paper-equity curve (no real orders). Going live means swapping the logged order for a real exchange order via ccxt. Recommended path: paper-trade forward 1–3 months, then trade unlevered at minimum size before scaling toward the 2–3x ceiling.

8. Risks & Caveats

- **In-sample optimism.** The Sharpe figures (3–4 with the trend filter) are upper bounds; real trading (slippage, funding, flip costs) will be lower. The shape of the improvement is trustworthy; the magnitude is not a forecast.
- **Regime dependence.** This is a momentum strategy. It thrives in trending markets and struggles in chop; the recent 2024–2026 crypto regime has been comparatively soft for momentum.
- **Crypto-specific.** The edge does not transfer to efficient markets (e.g. EUR/USD). It relies on crypto's trending, retail-driven inefficiency, which is not guaranteed to persist.
- **Not yet forward-tested.** All results are historical. Live forward paper-trading is the final, uncontaminated test and must be completed before committing real capital.
- **Leverage risk.** On crypto perpetuals, high leverage carries liquidation and funding risk worse than the idealized sweep suggests. Start unlevered.

This document summarizes a research strategy for informational purposes only. It is not financial advice and carries no guarantee of future performance. Trade only capital you can afford to lose.